Apparent Complexity of Quantized Diffusion, and Class-4 Structures Against the Boundary Strata of a Box

Miguel E. Pérez Ibarra* Jaziel Flores Rodríguez†

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Code and figures: https://github.com/MiguelPerezI/heat-parity-ca

Abstract

Part I. Colouring the cells of a finite-difference heat simulation by the parity of $\lfloor q\,u \rfloor$ produces black-and-white frames whose apparent complexity rises from nothing, peaks in a lattice of symmetric concentric islands, and dies. We show that these frames are the contour map of the temperature at spacing $1/q$ with alternating fill, that the rise is aliasing, and that the islands are the replicated centres of an undersampled Fresnel zone plate. Three elementary results make this exact: the band count and the total variation are nonincreasing under Neumann heat flow on a box; the phase-averaged perimeter of the sampled pattern equals a sum over grid edges of a triangle wave of the edge jump, $\sum_e \operatorname{dist}(q|\Delta_e u|, 2\mathbb{Z})$, and the discrete total variation is nonincreasing, so every increase of the perimeter is produced by edges whose jump exceeds one quantum; and the block entropy of the pattern has two non-commuting limits, 0 as the grid is refined and maximal as the quantum is refined. Part II. We then move the modulus from the display into the dynamics, which yields Wolfram's continuous cellular automata, and ask how their class-4 localized structures interact with the 3^n boundary strata of a box when the walls are reflecting, implemented by the ternary-cube rule.

Part I
Quantized diffusion: what the pictures are

1 The observation

The two-dimensional heat equation $u_t = \kappa \Delta u$ was integrated on a 64×64 grid with an explicit scheme, starting from a hot square ($u = 100$ on a 24×24 block, 0 elsewhere), and displayed as a height field in which each facet is coloured white or black according to the parity of $\lfloor 10\,u \rfloor$. The frames (Figure 1) begin uniformly black, erupt into a fine noise-like texture around the square, organize into a square-symmetric lattice of concentric islands, and finally fade to a few rings and then to a single colour. The sequence looks like a cellular automaton with a rich transient, and the islands look like eddies. Neither is the case, and the point of Part I is to say precisely what they are.

Throughout, $\Omega = (0, L)^n$, u solves $u_t = \kappa \Delta u$ with homogeneous Neumann data, $q > 0$ is the quantization frequency (quantum $1/q$), $\theta \in [0, 1)$ is a phase, and the *pattern* is

$$b_\theta(x, t) = \lfloor q\,u(x, t) + \theta \rfloor \bmod 2. \tag{1}$$

*Affiliation and e-mail. (Placeholder.)
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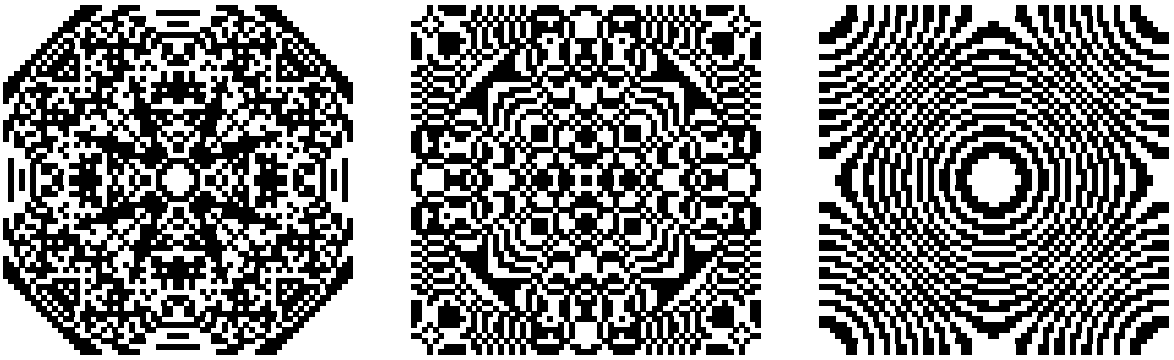


Figure 1: Parity of $\lfloor 10u \rfloor$ on the 64×64 grid at steps 150, 1000 and 2000 of the explicit scheme with $\Delta t = 2 \cdot 10^{-4}$, $\kappa = 0.2$, reflecting walls. The field has a single maximum throughout.

The grid version samples u at the N^n nodes of the explicit scheme with reflected ghost values (the ternary-cube rule of [1] with zero flux), under $\sum_m r_m \le \frac{1}{2}$. We write E for the set of grid edges and $\Delta_e u$ for the jump of u across the edge e.

So (1) is the contour map of u with isotherms at the levels $(k-\theta)/q$ and alternating fill. The pattern is an *observable* of u, not a state: two fields with the same pattern evolve to different patterns, so no local rule on black and white cells reproduces the frames, and by the usual definition (a map on configurations over a finite alphabet given by a local rule) this is not a cellular automaton.

2 What is monotone in the continuum

Theorem 2.1. *Let u solve the Neumann heat equation on the box Ω.*

1. *The band count $B_q(t) = \#\{k : (k-\theta)/q \in [\min u(\cdot,t), \max u(\cdot,t)]\}$ is nonincreasing for every θ.*
2. *The total variation $\mathrm{TV}(t) = \int_\Omega |\nabla u(x,t)|\, dx$ is nonincreasing.*
3. *The phase-averaged total isotherm length equals the total variation exactly:*

$$\int_0^1 \sum_{k\in\mathbb{Z}} \mathcal{H}^{n-1}\big(\{u(\cdot,t) = (k-\theta)/q\}\big)\, d\theta \;=\; q\ \mathrm{TV}(t), \tag{2}$$

and is therefore nonincreasing.

Proof. (1) is the maximum principle: $\max u$ is nonincreasing and $\min u$ nondecreasing. (2) Reflect u evenly across every face of the box; the result $\tilde{u}$ is a $2L$-periodic solution of the heat equation on the torus $\mathbb{T} = (\mathbb{R}/2L\mathbb{Z})^n$, smooth for $t > 0$. Each component $\partial_m \tilde{u}$ is again a periodic heat solution, so $\nabla\tilde{u}(t) = K_{t-s} * \nabla\tilde{u}(s)$ for $t > s$, where K is the periodic heat kernel, $K \ge 0$, $\int K = 1$. Convexity of the Euclidean norm gives $|\nabla\tilde{u}(t)| \le K_{t-s} * |\nabla\tilde{u}(s)|$ pointwise; integrating over $\mathbb{T}$ and dividing by 2^n gives $\mathrm{TV}(t) \le \mathrm{TV}(s)$. (3) By the coarea formula, $\int_\Omega |\nabla u| = \int_{\mathbb{R}} \mathcal{H}^{n-1}(\{u = s\})\, ds$; the left side of (2) is $\int_{\mathbb{R}} \mathcal{H}^{n-1}(\{u = s\})\, q\, ds$ after the substitution $s = (k-\theta)/q$, $\theta \in [0,1)$, $k \in \mathbb{Z}$. □

For a fixed phase the sum in (2) is a Riemann sum of the coarea integral, hence within $o(1)$ (as $q \to \infty$) of a nonincreasing function. In the continuum the quantized contour map has no rise of complexity, for any quantum, in the phase-averaged sense.

3 The exact perimeter formula on the grid

Let P_h be the number of grid edges whose two cells differ in colour, and $C_h = \sum_{e\in E} |\lfloor qu_a+\theta\rfloor - \lfloor qu_b+\theta\rfloor|$ the number of isotherm crossings, where a, b are the ends of e. Let $\mathrm{TV}_h = \sum_{e\in E} |\Delta_e u|$ and let $\varphi(x) = \mathrm{dist}(x, 2\mathbb{Z})$ be the triangle wave of period 2 with $\varphi(x) = x$ on $[0, 1]$.

Theorem 3.1. *1. (Discrete TVD.) For each axis m, $\sum_{\mathbf{i}} |u_{\mathbf{i}+\mathbf{e}_m} - u_{\mathbf{i}}|$ is nonincreasing under the explicit scheme with reflected ghosts and $\sum_m r_m \le \frac{1}{2}$. Hence TV_h is nonincreasing.*

2. (Random phase.) With θ uniform on $[0, 1)$,

$$\mathbb{E}_\theta C_h = q\ \mathrm{TV}_h, \qquad \mathbb{E}_\theta P_h = \sum_{e\in E} \varphi\big(q|\Delta_e u|\big), \tag{3}$$

both exactly.

3. Consequently: if $q|\Delta_e u| \le 1$ on every edge then $\mathbb{E}_\theta P_h = q\,\mathrm{TV}_h$ and is nonincreasing; always $\mathbb{E}_\theta P_h \le \min(q\,\mathrm{TV}_h, |E|)$; every increase of $\mathbb{E}_\theta P_h$ in time is produced by edges whose jump exceeds one quantum and is decreasing through a folded branch of φ; and $\mathbb{E}_\theta P_h$ is nonincreasing from the first time t^* at which $\max_e q|\Delta_e u| \le 1$.

Proof. (1) On the reflected periodic grid the scheme is a convex combination of translates, $u^{l+1} = \sum_j c_j\, \sigma_j u^l$ with $c_j \ge 0$, $\sum c_j = 1$, by the CFL condition. The axis difference $D_m u = \sigma_{\mathbf{e}_m} u - u$ satisfies the same recursion, and a convex combination of translates is an ℓ^1-contraction, so $\|D_m u^{l+1}\|_1 \le \|D_m u^l\|_1$. Restricting the periodic sum to the box changes it by the factor 2^n. (2) Under a uniform shift the number of levels $(k-\theta)/q$ inside an interval of length ℓ is $\lfloor q\ell\rfloor$ with probability $1 - \{q\ell\}$ and $\lfloor q\ell\rfloor + 1$ with probability $\{q\ell\}$. Its expectation is $q\ell$, which gives the first identity. The two ends of the edge differ in colour iff that number is odd; if $\lfloor q\ell\rfloor$ is even this has probability $\{q\ell\}$, otherwise $1 - \{q\ell\}$, and both equal $\varphi(q\ell)$. (3) follows from $\varphi(x) = x$ on $[0, 1]$, $\varphi \le \min(x, 1)$, and the fact that φ is increasing only on the branches $[2j-1, 2j]$, $j \ge 1$, which require $x > 1$. □

Theorem 3.1 explains the frames. Initially the jumps across the edge of the hot square are a thousand quanta, φ is folded a thousand times, and the expected perimeter per such edge is about $\frac{1}{2}$. As the front spreads the number of edges in the folded regime grows, so the perimeter grows, although $q\,\mathrm{TV}_h$ is falling all along. When the largest jump passes below one quantum, the perimeter becomes $q\,\mathrm{TV}_h$ itself and decays with it. Table 1 shows the identity (3) against a Monte-Carlo average over 200 phases on the setup of Figure 1, and the peak at the step where the largest jump crosses one quantum.

4 Entropy: the limits do not commute

Let $H_{q,h}(t)$ be the Shannon entropy of the empirical distribution of 2×2 blocks of b_θ on the grid of spacing h.

Theorem 4.1. *Fix $t > 0$.*

1. For fixed q, $\lim_{h\to 0} H_{q,h}(t) = 0$.

2. For fixed h, if $1, u(x_1, t), \dots, u(x_M, t)$ are linearly independent over $\mathbb{Q}$, where x_j are the grid nodes, then $(q\,u(x_j, t) \bmod 2)_j$ is equidistributed on $(\mathbb{R}/2\mathbb{Z})^M$ as $q \to \infty$, and the Cesàro limit over q of $H_{q,h}(t)$ is the maximum $\log_2 16 = 4$ bits.

step	$q\,\mathrm{TV}_h$	$P_h(\theta=0)$	$\overline{P_h}$ (MC)	$\sum_e \varphi(q\lvert\Delta_e u\rvert)$	$\max_e q\lvert\Delta_e u\rvert$
20	95999.7	1136	1256.2	1257.3	155.95
150	87103.5	3600	3613.3	3610.7	51.75
400	59437.5	4032	4004.1	4004.7	22.19
1000	22542.9	4232	4285.2	4286.4	5.75
1500	10006.7	3560	3552.3	3555.3	2.22
2000	4435.4	4432	4436.8	4435.4	0.92
2500	1964.6	1976	1963.5	1964.6	0.39
3000	870.0	896	867.3	870.0	0.17
4000	170.5	184	172.9	170.5	0.03

Table 1: The random-phase identity (3) on the 64×64 run of Figure 1, $q = 10$. The discrete total variation never increased; the perimeter peaks where the largest jump crosses one quantum; from then on $\overline{P_h} = q\,\mathrm{TV}_h$ to rounding.

Proof. (1) For $t > 0$ the isotherms are smooth curves of finite total length $\Lambda_q(t)$; a 2×2 block is non-uniform only if an isotherm passes within distance $2h$ of it, so the fraction of non-uniform blocks is $O(h\Lambda_q(t))$, and the entropy of a distribution concentrated on one symbol with mass $1 - O(h)$ is $O(h \log(1/h))$. (2) is Weyl's equidistribution theorem (Kronecker–Weyl) for the sequence $q \mapsto (qu(x_j, t))_j$ on the torus, under the stated rational independence; the uniform distribution on $\{0,1\}^4$ maximizes the block entropy. □

The square-symmetric initial data of the experiment violate the independence in (2), since $u(x_j) = u(\sigma x_j)$ for the symmetries σ of the square, and the achievable entropy is reduced by the orbit structure. That is why the aliased frames look like symmetric designs rather than noise. Measured on the frames of Figure 1, the block entropy reaches 3.99 bits at step 400; this is regime (2), a statement about the sampling, not about the heat equation.

5 The islands are moiré

Where u is locally linear, $\lfloor q(a \cdot x) + \theta \rfloor \bmod 2$ on the lattice is a two-colour digital plane, whose one-dimensional sections are Sturmian words; these are the striped regions. Where u is locally quadratic, near the maximum and in the tails, the pattern is a sampled Fresnel zone plate. An undersampled zone plate replicates its centre: neighbouring cells agree in colour wherever $q\,h\,\partial_m u$ is near an even integer, so fake centres appear along the curves $\{qh\,\partial_m u \in 2\mathbb{Z}\}$, $m = 1, 2$. Figure 2 shows the effect in its purest form: the state at $t = 0.03$, when the field still carries 832 isotherms and the gradient at the edge of the square is about forty quanta per cell even on a 256^2 grid. The true contour map would be a set of tightly nested rounded squares; what the lattice sees instead is a central zone plate surrounded by a ring of replica centres, each a small zone plate of its own, arranged on the curves where $qh\,\partial_m u$ passes through even integers, with the quasi-periodic stripes of digital planes between them. Every feature of this picture is a property of the sampling: the field has one maximum, and its isotherms never cross. Figure 3 shows the later state $t = 0.4$ at 64 and 256 cells with the same quantum: the island lattice is present at 64 and absent at 256, where the frame is an ordinary contour map. Table 2 gives the measures. With the quantum scaled as $1/N$ instead, the measures are constant in N (perimeter per cell 0.879, 0.894, 0.893): the texture depends on the single ratio $qh|\nabla u|$ and is scale-free undersampling.

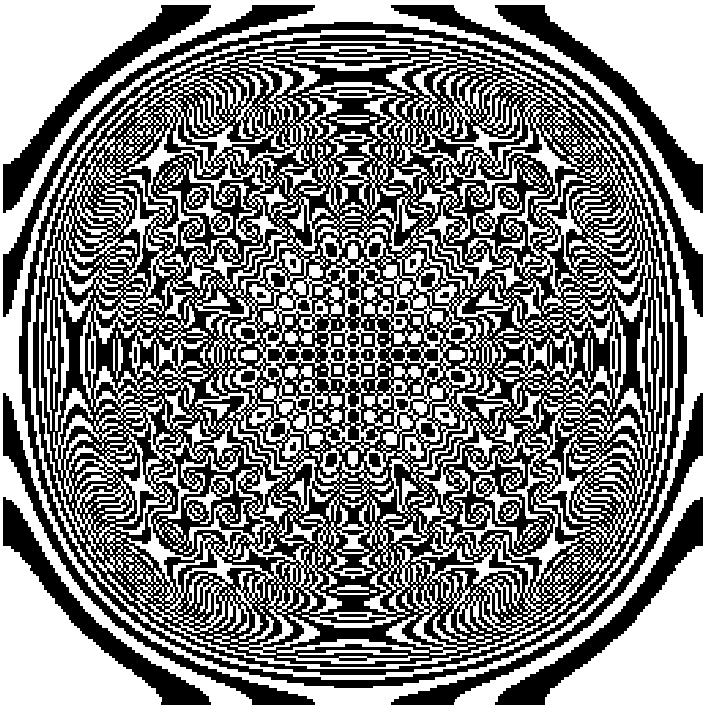


Figure 2: The parity pattern at $t = 0.03$ on a 256^2 grid, $q = 10$, 832 isotherms. An undersampled Fresnel zone plate: the central rings are the real contour map of the peak, the surrounding lattice of small concentric centres is moiré, replicas created where the local fringe frequency matches a multiple of the sampling frequency, and the striped sectors between them are digital planes. The figure is vector graphics; zooming in shows the individual cells.

N	bands	perimeter/N^2	$H_{2\times2}$ (bits)	components
64	34	1.082	3.93	905
128	36	0.564	3.45	88
256	37	0.287	2.61	92

Table 2: Resolution test at $t = 0.4$, $q = 10$ fixed. Perimeter per cell halves with each doubling, as a fixed set of smooth curves does; the islands collapse into the real rings.

6 What is not claimed, and what the lesson is

The number of components of the pattern, and the number of local maxima of u, are not monotone in general in two dimensions [3]; nor is the entropy of the value histogram of u. There are no eddies: for $t > 0$ the field is analytic with smooth nested isotherms. The lesson of Part I is methodological. Perimeter, block entropy and component counts of a thresholded simulation measure the sampling as much as the dynamics; the phase-averaging identity (3) is a diagnostic, since a phase-averaged perimeter exceeding $q\,\mathrm{TV}_h$ is aliasing by Theorem 3.1(3). Real complexity requires the nonlinearity inside the dynamics, which is the subject of Part II.

A remark on functional information. Wong et al. [13] have proposed a law of increasing functional information: the functional information of a system increases when many configurations of the system undergo selection for one or more functions. Functional information, in the sense of Szostak and Hazen [14, 15], is $I(E_x) = -\log_2 F(E_x)$, where $F(E_x)$ is the fraction of all configurations that perform a given function to degree at least E_x; it is defined only relative to



Figure 3: Left: 64×64 grid at $t = 0.4$ ($q = 10$). Right: the same state on a 256×256 grid with the same quantum. The islands of the left frame are replicated zone-plate centres; the right frame is the contour map of the field, with its mirror images in the four corners produced by the reflecting walls.

a function, and the proposed law makes selection, not mere dynamics, the source of its growth. Part I is a clean example of why the restriction to a function is necessary. The quantized heat frames have no function and undergo no selection; every quantity intrinsic to the field, the band count, the total variation, the gradient, decreases monotonically (Theorems 2.1 and 3.1); and yet the function-free complexity proxies that are commonly read as evidence of "ratcheting" complexity, perimeter, block entropy, component count, rise from zero to near their maxima before they fall. The rise is produced entirely by the observer's choices of quantum and grid, and Theorem 4.1 shows that these choices can make the block entropy anything between zero and its maximum. The random-phase identity (3) is the diagnostic that separates the two cases: a phase-averaged perimeter above $q\,\mathrm{TV}_h$ is sampling, not structure. The automata of Part II are a second data point of the same kind: they sustain persistent localized structures, which is what the word complexity usually refers to, but nothing in them is selected for anything, and the boundary strata act as mirrors, not as selectors. Whether a system with persistent structures but without a selected function should count as evolving in the sense of [13] is a question that framework answers in the negative by construction, and our examples suggest that this is the right answer.

Part II
Class-4 structures against the boundary strata

7 Moving the modulus into the dynamics

Wolfram's continuous cellular automata [2, ch. 4, 6] replace the display rule of Part I by a state rule:

$$u_{\mathbf{i}}^{l+1} = \mathrm{frac}\Big(u_{\mathbf{i}}^{l} + \sum_{m=1}^{n} r_m\, \tilde{\delta}^{2}_{m,\tau_m(\mathbf{i})} u_{\mathbf{i}}^{l} + a\Big), \tag{4}$$

with $r_m = 1/(2n+1)$ this is "the mean of the $(2n+1)$-point neighbourhood plus a, modulo 1". The operator $\tilde{\delta}^{2}_{m,\tau_m}$ is the per-axis second difference of the ternary-cube rule [1]: on a free digit it is the ordinary central difference, on a wall digit it uses the reflected ghost value, so the box has reflecting walls and the 3^n strata of the grid are exactly the places where the rule differs. In one dimension Wolfram reports class-4 behaviour, with discrete localized structures on a periodic background, for $a = 0.398$, $a = 0.4$, and for $a = 0.5$ with the neighbours weighted by

1.13 before averaging [2, p. 244].

8 Questions

A localized structure that reaches a wall of codimension one meets a cell whose neighbourhood is its own mirror image. In two dimensions it can also meet an edge where two mirrors cross, and in three dimensions a corner where three cross. We ask:

1. Do structures reflect, annihilate, or stick at a wall? Does the wall itself nucleate structures?
2. Is the time-averaged activity near a wall higher, lower or equal to the bulk value, and how does it depend on the codimension of the stratum (wall, edge, corner)?
3. Are there boundary-bound structures that exist only on a stratum, and does their class differ from the bulk class?

9 The reflection principle

The first result is exact and settles the structure of the problem. Write $\mathcal{R}_m$ for the even reflection of a configuration across the wall nodes of axis m, which turns the box grid $\{0, \dots, N_m - 1\}$ into the periodic grid of length $2(N_m - 1)$, and $\mathcal{R} = \mathcal{R}_1 \cdots \mathcal{R}_n$ for the reflection into the torus $\mathbb{T}_h = \prod_m \mathbb{Z}/2(N_m - 1)\mathbb{Z}$.

Theorem 9.1 (reflection principle)**.** *Let F_{box} be the rule (4) on the box with ternary-cube mirror ghosts, and F_{per} the same rule on $\mathbb{T}_h$ with periodic wrap. Then*

$$\mathcal{R} \circ F_{\mathrm{box}} = F_{\mathrm{per}} \circ \mathcal{R},$$

and $\mathcal{R}$ maps the box configurations bijectively onto the reflection-symmetric configurations of $\mathbb{T}_h$, which form an invariant set of F_{per}. The same holds for any rule whose per-axis stencil is symmetric under $i \mapsto -i$, in particular for the weighted means of [2, p. 244], and for any pointwise post-processing such as frac *or rounding.*

Proof. At a node with free digit $\tau_m = 1$ the two rules read the same neighbours. At a wall node the mirror ghost $u_{-1} := u_1$ is exactly the value that the reflected configuration carries at the position -1, so the stencil sums agree; symmetry of the stencil makes the reflected configuration's update at $-i$ equal to its update at i, so symmetry is preserved. Pointwise maps commute with everything. □

Numerically the identity holds to the last digit: on 400 cells for 3000 steps in one dimension and on 64^2 cells for 1000 steps in two dimensions, the maximum difference between the box run and the doubled periodic run was 0 (script `reflection_principle.py`).

The consequences organize Part II. A structure that hits a wall of codimension one collides head-on with its own mirror image; what the wall does to it is the outcome of a symmetric two-body collision. A structure that reaches an edge of a three-dimensional box, or a corner of a two-dimensional one, meets the images under two commuting reflections, a symmetric four-body collision; a corner of a three-dimensional box is an eight-body collision. The stratum of codimension c is the fixed set of c commuting reflections, and the activity there is constrained accordingly: the wall node is its own image, so the difference across the wall is identically zero, and the boundary stratum is *quieter* than the bulk by construction, not by dynamics. Finally, nothing can live on a wall that does not also exist in the periodic system as a symmetric configuration, so boundary-bound structures are exactly the mirror-symmetric localized structures of the bulk rule pinned at the symmetry axis.

10 One dimension

We take $a = 0.398$, $r = \frac{1}{3}$ (the three-cell mean), random initial data, values rounded to nine decimals after each step to emulate exact arithmetic, and the activity indicator $\chi_{\mathbf{i}} = [\mathrm{dist}(u_{\mathbf{i}} - u_{\mathbf{i}+1}, \mathbb{Z}) > 0.05]$, which is Wolfram's difference display thresholded. Figure 4 shows the space-time diagrams with periodic and with reflecting walls, and a magnified view of the wall.

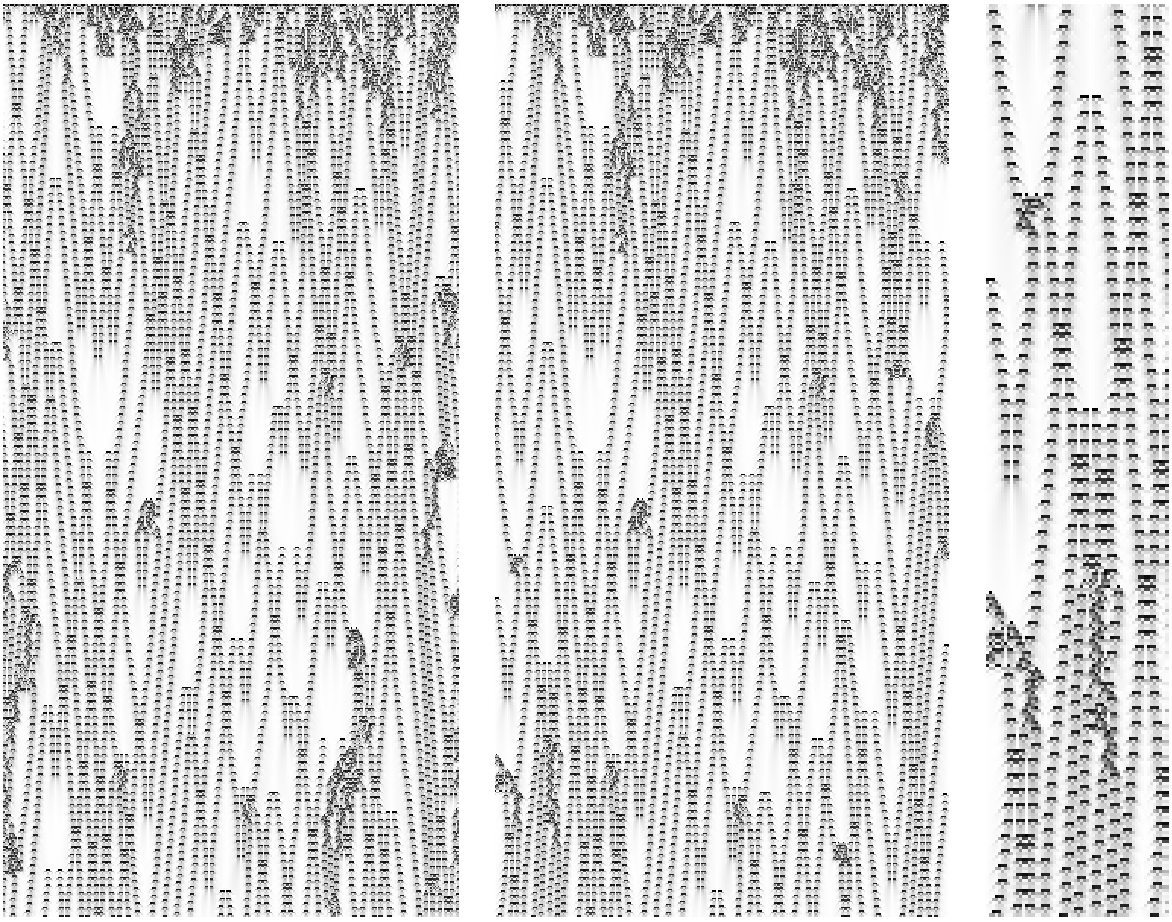


Figure 4: Class 4 in one dimension: $a = 0.398$, three-cell mean plus a modulo 1, 300 cells, 600 steps downward from random data, same seed, difference display (black where neighbours differ). Left: periodic. Middle: mirror walls at both sides. Right: the 60 cells nearest the left wall during steps 300–600, magnified. Localized structures move at fixed speeds through a periodic background, cross or scatter, and at the wall return as their mirror images, forming the V shapes predicted by the reflection principle.

The bulk activity fraction, averaged over four seeds and 2500 steps after a 500-step transient, is 0.27 with periodic boundaries and 0.28 with walls: intermediate between the extinct and the saturated value, as class 4 requires. The profile against distance to the wall is flat in the bulk and drops only in the wall layer: the wall node has activity 0.09 and its neighbour 0.23, after which the bulk value is reached. The ten cells nearest the walls carry 0.90 of the bulk activity. This is the quiet layer predicted by Theorem 9.1: the wall node cannot differ from its own image, and its neighbour sits in a symmetric three-cell neighbourhood. No structure was seen to stick at the wall, and the wall did not nucleate structures beyond those arriving from the bulk, consistent with the collision picture: a head-on symmetric collision in this rule returns the pair.

11 Two and three dimensions

Scan. In two dimensions the rule is the five-point mean plus a modulo 1. A scan of a in steps of 0.01 on a 128^2 torus from random data (script `scan2d.py`) gives a late activity fraction that varies *continuously* with a, from 0.29 near $a = 0.01$ to 0.92 on the plateaus $a \in [0.34, 0.43]$ and $[0.57, 0.66]$, with dips to 0.43–0.45 near $a = \frac{1}{3}, \frac{1}{2}, \frac{2}{3}$. There are no sharp class-4 windows of the one-dimensional kind, and the one-dimensional class-4 value $a = 0.398$ is fully chaotic in two dimensions (activity 0.92, persistence of active cells equal to their density). Space-time cuts (Figure 5) show three regimes: near $a = \frac{1}{3}$ and $\frac{2}{3}$ the background is a period-three cycle and the dynamics consists of slowly wandering *domain walls* between its three phases; near $a = \frac{1}{2}$ one sees intermittent short-lived spots on a period-two background; on the plateaus, spatiotemporal chaos. In this family the two-dimensional analogue of the one-dimensional gliders is therefore a codimension-one defect, the phase wall, and the natural question becomes how phase walls and intermittent spots meet the box.

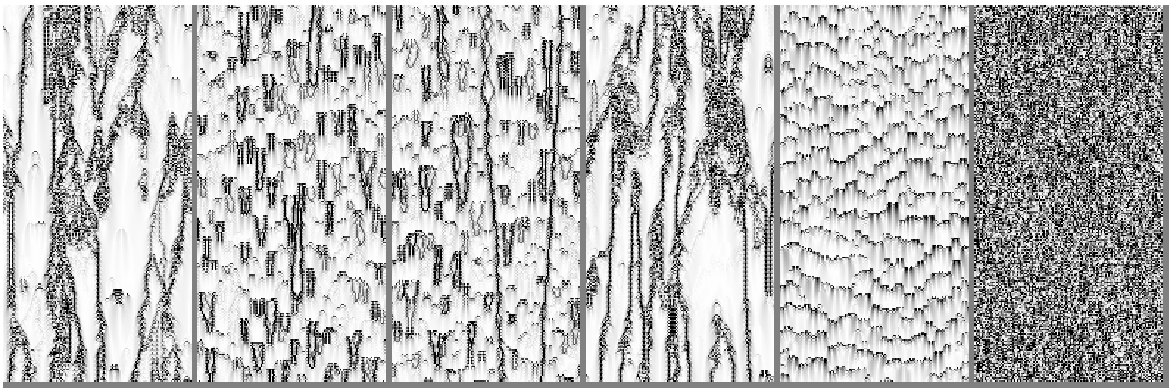


Figure 5: Space-time cuts (middle row of a 128^2 torus against the last 256 of 1200 steps, difference display) for $a = 0.33$, 0.49, 0.51, 0.67, 0.02, 0.398. Wandering phase walls at 0.33 and 0.67, intermittent spots at 0.49 and 0.51, a regular lattice at 0.02, chaos at 0.398.

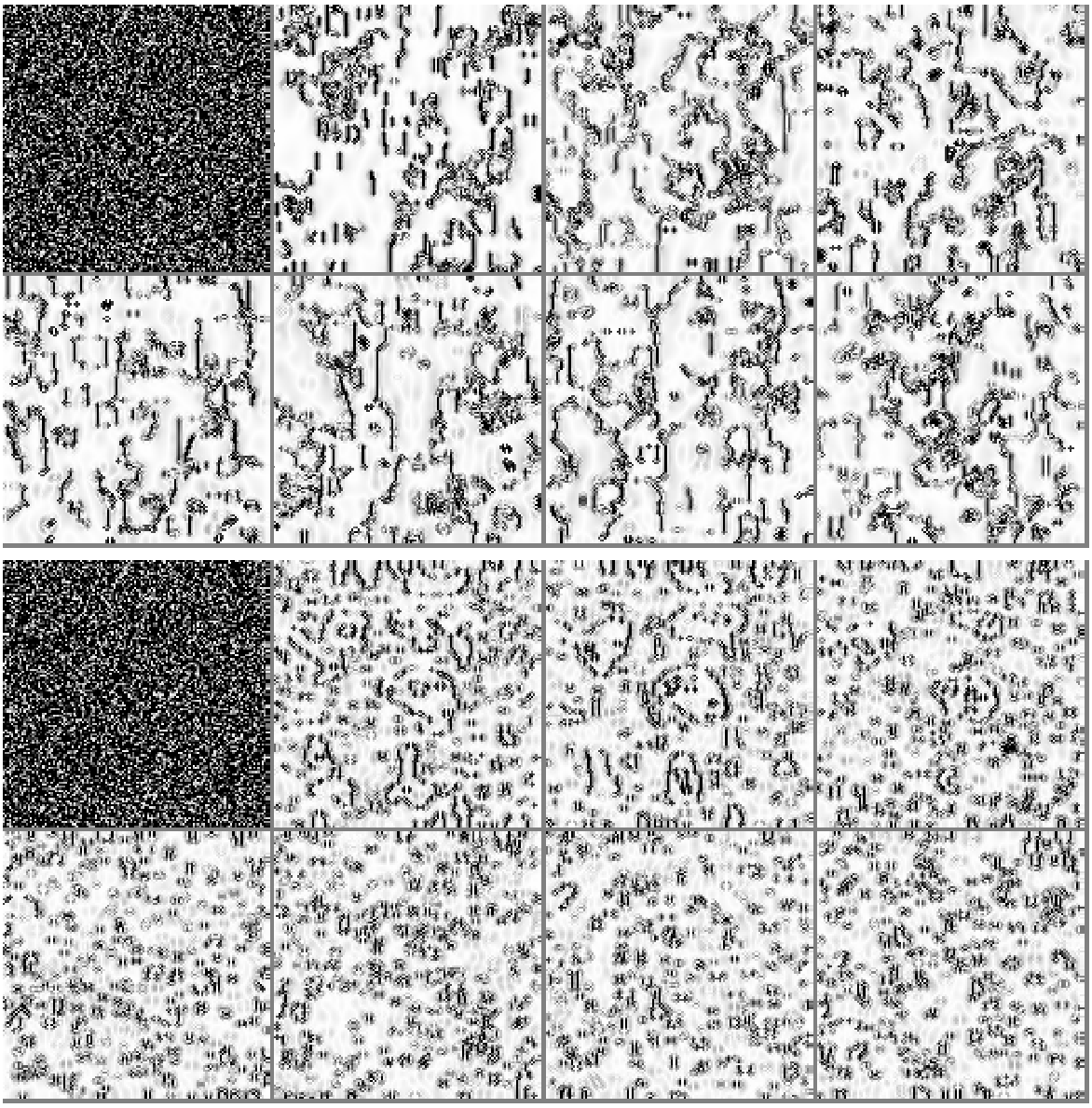


Figure 6: The 128^2 box with mirror walls, difference display, at steps $0, 50, 100, 200$ (top row of each strip) and $400, 600, 800, 1200$ (bottom row). Upper strip, $a = 0.33$: random data collapse within fifty steps into a period-three background crossed by phase walls, which then wander, join and pinch off; walls parallel to a side are cleared from a margin a few cells wide, walls perpendicular to it end on it. Lower strip, $a = 0.49$: intermittent spots on a period-two background, born and dying everywhere except at the corners.

Activity per stratum. We run the box with mirror walls, classify every node by the codimension of its stratum in the ternary cube [1] (0 interior, 1 wall, 2 edge or two-dimensional corner, 3 three-dimensional corner), and record the time-averaged activity indicator, now defined as the maximum over all existing neighbour differences of $\operatorname{dist}(u_{\mathbf{i}} - u_{\mathbf{j}}, \mathbb{Z}) > 0.05$. Grids are 128^2 for 1500 steps and 32^3 for 800 steps, averaged after a transient of one third of the run (script `strata.py`). Table 3 gives the results.

Four observations. First, in the chaotic regime the activity decreases monotonically with the codimension of the stratum, in both dimensions, and the depth of the boundary layer is one cell: at distance one from the nearest wall the two-dimensional activity is already 0.981 against 0.989 in the bulk. This is the kinematic effect of Theorem 9.1: a node on a stratum of

regime	grid	codim 0	codim 1	codim 2	codim 3
chaos, $a = 0.398$	128^2	0.989	0.948	0.739	
chaos, $a = 0.398$	32^3	0.997	0.993	0.981	0.927
intermittent, $a = 0.49$	128^2	0.561	0.503	0.190	
intermittent, $a = 0.49$	32^3	0.475	0.494	0.500	0.334
phase walls, $a = 0.33$	128^2	0.491	0.349	0.207	
chaos (3-D rule), $a = 0.33$	32^3	0.999	0.998	0.988	0.898

Table 3: Time-averaged activity fraction per boundary stratum of the box with mirror walls. One seed per row; the three-dimensional values fluctuate by about ± 0.03 between distances in the bulk.

codimension c has c of its neighbour pairs identified by symmetry, so it has fewer independent neighbours to differ from, and the effect compounds with c. Second, in the intermittent regime the corner is strongly suppressed in both dimensions (0.19 against 0.56 in two dimensions, 0.33 against 0.48 in three), while the three-dimensional faces and edges are at the bulk value within the noise. So in two dimensions the spots avoid the corners but not the walls, and in three dimensions only the eight corner nodes are special; the twelve edges, which are the analogue of the two-dimensional corners, are not. Third, the phase-wall regime is the one in which the box is felt at a distance: the activity profile against the distance to the nearest wall is 0.35, 0.40, 0.44, 0.47, 0.48, 0.49 for distances 0 to 5, a boundary layer about five cells deep, against one cell in the other regimes. The reflection principle says why: a phase wall approaching a mirror meets a phase wall of opposite orientation, its own image, and the pair annihilates when the strip between them collapses, so a region near the mirror is cleared of walls parallel to it; walls perpendicular to the mirror end on it, and are the activity that remains at distance zero. Fourth, the same $a = 0.33$ is chaotic under the seven-point rule in three dimensions, with the one-cell layer of the chaotic regime, which shows that the regime, not the constant, is what the strata respond to.

What this does and does not show. Within the mean-plus-constant family, the strata of the ternary cube are visible as the reflection principle predicts: a quiet layer whose strength grows with the codimension, strongest at the corners, one cell deep where the bulk is chaotic or intermittent and several cells deep where the bulk is organized into phase domains, because there the mirror annihilates its own image's walls. We did not find two-dimensional class-4 structures of the glider type in this family, so the question of how a glider reflects off an edge or a corner remains to be asked of a rule that has gliders, for instance Lenia [12], where the large convolution kernel makes the boundary strata $(2s + 1)^n$ layers thick in the sense of the ternary-cube note [1]. The reflection principle applies there verbatim.

Code

All scripts, the figures and the source of this article are public at `https://github.com/MiguelPerezI/heat-parity-ca`. The scripts are in `code/`: `analyze.py`, `restest.py` and `phase_theorem.py` for Part I; `contca.py`, `scan1d.py`, `nks_repro.py`, `wall1d.py`, `reflection_principle.py`, `scan2d.py`, `cuts2d.py` and `strata.py` for Part II; `view1d.py` and `filmstrip2d.py` produce the figures of Sections 10 and 11, and `view2d.py` is a live viewer of the box (`python3.12 view2d.py 0.33 160 mirror`).

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